Juan Becerra

Software Engineer – Gameplay, AI Systems & ML Pipelines

Seattle, WA | (208) 550-9191 | becerrajuan007@gmail.com | hoowan.dev
linkedin.com/in/hoowan | github.com/hoowan-dev

Experience

Electronic Arts – Lifestyle **2025 – Present**
Software Engineer – Tools & Tech-Art *Kirkland, WA*

- Engineered a high-throughput Unreal Engine C++ plugin exporting structured runtime character trajectory, pose, and velocity datasets to train and populate motion-matching query databases.
- Prototyped and integrated ML-assisted pipelines into authoring workflows (leveraging Python, PyTorch, and ONNX runtime), enabling automated asset tagging and procedural pose generation without disrupting designer or artist control.
- Architected an extensible Maya/Python animation stylization tool and data processing pipeline adopted across multiple internal studios, reducing redundant manual keyframing workflows by over 70%.

Electronic Arts – Maxis Studios **2021 – 2025**
Software Engineer – Gameplay & Animation *Seattle, WA (Remote)*

- Architected a multi-threaded asynchronous path-planning and agent navigation API in UE5/C++ for *Project Rene* (*The Sims*), enabling deterministic crowd routing and real-time obstacle avoidance within strict frame budgets (<1.5ms).
- Partnered directly with gameplay designers and technical animators to author in-editor tuning tools and visual debuggers, accelerating iteration cycles and authoring loops for complex agent behavior trees.
- Profiled and optimized runtime compute, memory, and frame budgets across character evaluation graphs and spatial query sweeps, eliminating hitches during dense multi-agent simulation scenarios.
- Implemented core locomotion and runtime animation systems in UE5 (motion-matching, distance matching, custom IK solvers), delivering fluid, responsive agent behaviors for next-gen simulation gameplay.

Lionbridge **2020 – 2021**
Senior Games Test Engineer *Boise, ID*

- Led a team of QA testers performing combinatorial and regression testing for live-service AAA titles, authoring technical repro steps and engine-level diagnostics in JIRA to accelerate developer triage.

Projects

Hoowan Game Engine | *C++, OpenGL, EnTT, CMake* | GitHub **Nov. 2020**

- Engineered a custom C++ game engine featuring an OpenGL rendering submission pipeline, input event polling, and 2D collision detection and correction systems.
- Integrated an EnTT-based entity-component system (ECS) and constructed a companion desktop editor application for rapid scene composition and entity management.

DoomSlop | *Three.js, Web Audio API, Vite, JavaScript* | Live Demo | GitHub **Sep. 2026**

- Built a 3D browser first-person shooter featuring hitscan raycasting, synthesized Web Audio sound effects, custom canvas radar minimap, and automated CI/CD deployments via GitHub Pages.

Technical Skills

Languages: C++ (Modern, Multi-threading), Python, SQL, Rust, C#, GLSL/HLSL, Java, JavaScript
Tools & ML Pipelines: PyTorch/ONNX, Unreal Plugin Development, DCC Pipeline Automation, In-Editor Tooling, Maya API, PySide/Qt
Gameplay & Systems: Async Navigation/Pathfinding, State Machines, Locomotion & Character Physics, Animation Runtime (IK, Motion-Matching)
Game Engines & APIs: Unreal Engine 5 (Source, Mass/Crowd, Motion Matching, GAS), Unity, Godot, Vulkan, OpenGL
Infrastructure & Tools: Perforce (P4), Git, Docker, Linux, AWS, CI/CD, CMake, Visual Studio, JIRA

Education & Continuing Studies

Boise State University **2017 – 2021**
Bachelor of Science in Computer Science *Boise, ID*

University of Washington (Continuing Education) **2026 – Present**
Graduate Coursework in Machine Learning & Deep Learning (Non-Degree, In Progress) *Seattle, WA*